

Public Table Datasets for LLMs & Table RAG

Personal research notes on publicly available datasets for Table RAG, Table Question Answering (Table QA), Text-to-SQL, and AI systems.

Goal

The purpose of this document is to collect useful table-centric datasets that can be used for building and evaluating:

- Table RAG
- Table Question Answering
- Text-to-SQL
- AI Agents
- Fact Verification
- Enterprise Search
- Financial AI

Rather than keeping random bookmarks, this is a single place to compare datasets and understand where each one fits.

Dataset Categories

- Table Question Answering (Table QA)
 - Text-to-SQL
 - Fact Verification
 - Hybrid Table + Text QA
 - Enterprise Databases
 - Financial Datasets
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Useful Resources

Resource	Link
Hugging Face Datasets	https://huggingface.co/datasets
Papers With Code	https://paperswithcode.com/datasets
Kaggle	https://www.kaggle.com/datasets
UCI ML Repository	https://archive.ics.uci.edu/ml
OpenML	https://www.openml.org
Google Dataset Search	https://datasetsearch.research.google.com

Recommended Datasets

Dataset	Best For	Link
WikiTableQuestions	Table QA	https://github.com/ppasupat/WikiTableQuestions
WikiSQL	Beginner Text-to-SQL	https://github.com/salesforce/WikiSQL
Spider	Advanced Text-to-SQL	https://yale-lily.github.io/spider
BIRD	Enterprise Text-to-SQL	https://bird-bench.github.io
TabFact	Fact Verification	https://tabfact.github.io
HybridQA	Table + Text QA	https://github.com/wenhuchen/HybridQA
FEVEROUS	Hybrid Retrieval	https://huggingface.co/datasets/feverous/feverous
SQA	Sequential Table QA	https://github.com/microsoft/SQA
ToTTo	Table-to-Text Generation	https://github.com/google-research-datasets/ToTTo
TURL	Table Understanding	https://github.com/sunlab-osu/TURL
FinQA	Financial QA	https://github.com/czyssrs/FinQA

Dataset	Best For	Link
TAT-QA	Financial Reasoning	https://nextplusplus.github.io/TAT-QA/

What I Learned

After looking through these datasets, a few things stood out.

- Most popular benchmarks are built using Wikipedia tables.
- Spider and BIRD are currently the most widely used benchmarks for Text-to-SQL.
- Enterprise systems usually need more than vector search; they combine SQL, metadata, and semantic retrieval.
- Financial datasets focus much more on numerical reasoning than general QA datasets.
- Hybrid datasets like HybridQA and FEVEROUS are closer to real-world applications because they require both tables and text.

Why Tables Are Challenging

Unlike normal documents, tables store information through rows, columns, and relationships.

Some common challenges are:

- Understanding schemas
- Numerical reasoning
- Aggregation
- Filtering
- Multi-table joins
- Preserving row-column relationships
- Handling large databases

This is why many modern systems combine SQL execution with retrieval instead of relying only on embeddings.

Project Ideas

Some interesting projects that could be built using these datasets:

- Build a Table RAG system over CSV or SQL databases.
 - Compare row-level vs table-level retrieval.
 - Fine-tune an LLM for Text-to-SQL using Spider.
 - Build an AI data analyst that converts natural language into SQL.
 - Create an enterprise chatbot that can answer questions from structured business data.
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Recommended Datasets by Use Case

Learning Table QA

- WikiTableQuestions
- SQA

Learning Text-to-SQL

- WikiSQL
- Spider

Enterprise AI

- BIRD
- Spider

Table RAG

- HybridQA

- FEVEROUS
- TabFact

Financial AI

- FinQA
 - TAT-QA
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Next Steps

Things I want to explore next:

- Table chunking strategies
 - Table embeddings
 - Hybrid SQL + Vector Search
 - Schema retrieval
 - Enterprise Table RAG architectures
 - Recent research papers on structured retrieval
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References

- Hugging Face — <https://huggingface.co/datasets>
 - Papers With Code — <https://paperswithcode.com/datasets>
 - ACL Anthology — <https://aclanthology.org>
 - arXiv — <https://arxiv.org>
 - Stanford NLP — <https://nlp.stanford.edu>
 - Google Research — <https://research.google>
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Status: Ongoing research. I'll continue updating this page as I discover new datasets, papers, and production approaches.